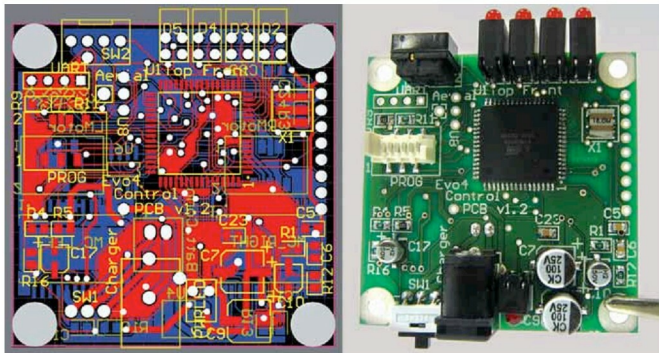


SMD Components

Calling for Investments in Manufacture

Currently, SMD component manufacturing capability in India is not worth talking about, the reason being it is a very capital-intensive industry and one needs to keep modernising the machinery to keep up with the technology. However, given the demand for SMD components growing at a fast pace, India can no longer afford to ignore local manufacture



Printed circuit board (PCB) layout created (left) on a computer and (right) the manufactured board using both through-hole and surface mounts components (Courtesy: Wikipedia)

UMA GUPTA

Miniaturisation continues to be the key trend driving the electronics industry. It contributes to reduction in cost while providing convenience, aesthetics and better quality in the final product. The advent of surface-mount technology (SMT) has given an impetus to this trend.

SMT enables construction of electronic circuits in which surface-mount components (SMCs) are mounted directly on the PCB. It offers major advantages such as automation, high-

speed assembly, miniaturisation, and better mechanical and electrical performance. Most of the new-age gadgets such as tablet PCs, laptops, display panels and mobile phones are manufactured using SMT.

SMT requires use of surface-mount devices (SMDs), which consist of passive components like chip resistors and capacitors, and active components like integrated circuits (ICs), diodes and transistors. Manufacturing of SMD components in India is almost negligible and therefore imports have been increasing rapidly.

Demand growing fast

The demand for SMD components in India is quite big and growing at a very fast pace. Chip resistors, chip capacitors, chip tantalum capacitors, SMD power inductors, SMD transistors and SMD diodes are selling in huge quantities.

Sujata Soparkar, managing director, Integrated Electronics Technology, cites two reasons for increase in the demand for SMD components: "First, due to miniaturisation, more and more leaded components are being replaced with SMD components. Second, the